Sustainability Economics: Hermit Economics

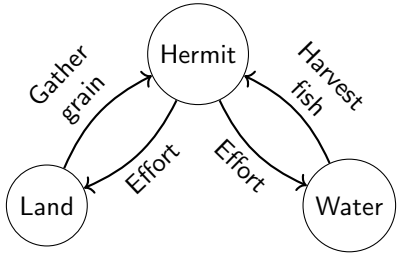
John Janmaat

October 27, 2026

Hermit System

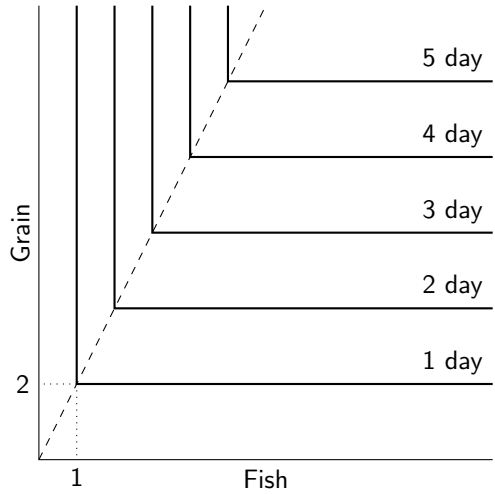
- Hermit,
  - The social element of the social ecological system.
  - Contributes labour and consumes the fruits of that labour.
- Field,
  - One ecological element of the system.
  - Produces grain, with labour as the main input.
- Lake.
  - One ecological element of the system.
  - Produces fish, with fishing effort (time spent fishing) as the main input.

Hermit System



Human Ecology

- Humans need food.
- Carbohydrates, fat and protein.
- Grain high in carbohydrates
- Fish provide fat and protein.
- Assume need fixed amounts of each.



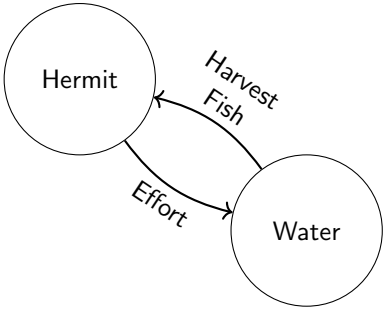
Production Functions

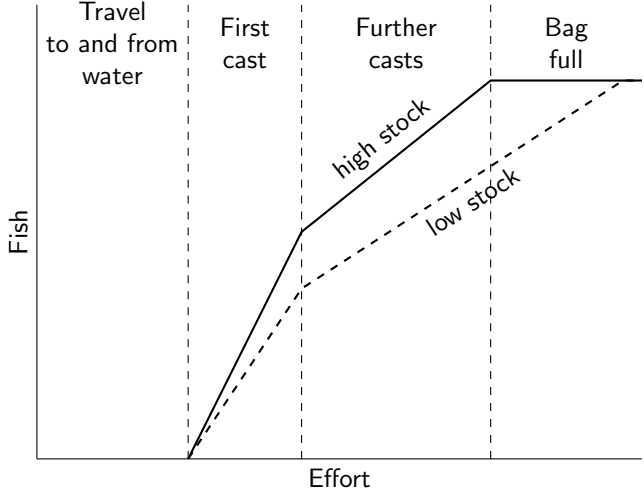
- A *technology* is a way that inputs are converted to outputs.
- A *production function* is a way of representing the technology.
  - The production function is a simplification.
  - The technology behaves the way that physics, chemistry, etc. require it to.
  - All models are wrong, some are useful.
- Three measures are commonly used to describe production at each input level.
  - *Total Product* is the total output produced given a particular input level.
  - *Average Product* is the total product divided by the input level.
  - *Marginal Product* is how much the total output changes if the level of input is changed by a small amount.

The Fish Subsystem

- Time path:
  1. Travel from village to water
  2. Cast net
  3. Collect fish from net until can't carry any more
  4. Travel back to village
  5. Exchange fish for grain
- Each step of time path requires effort.

The Fish Subsystem

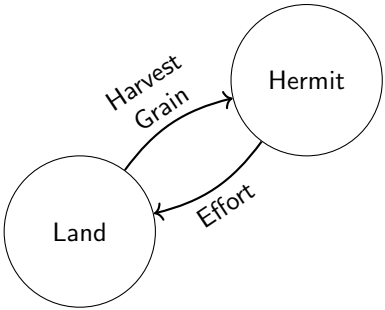


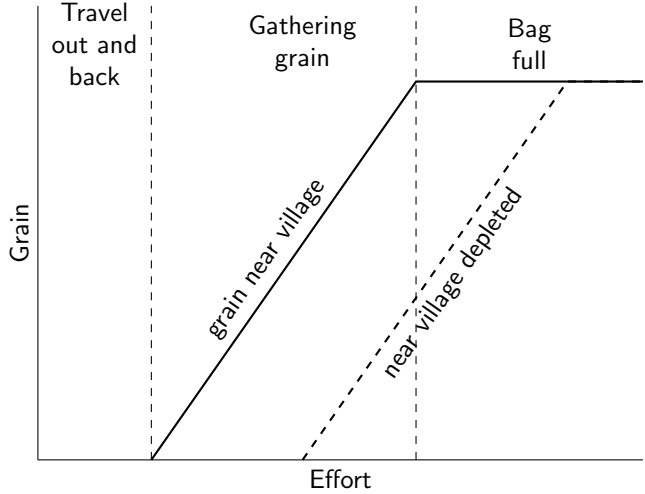


The Grain Subsystem

- Time path:
  1. Travel from village to closest patch of land with grain.
  2. Harvest grain from patch.
  3. Move to next patch and repeat until grain bucket full.
  4. Travel back to village
  5. Exchange fish for grain
- Each step of time path requires effort.

The Grain Subsystem





Total, Average and Marginal Product

- Total Product (TP) and Average Product (AP) are based on a level of input.

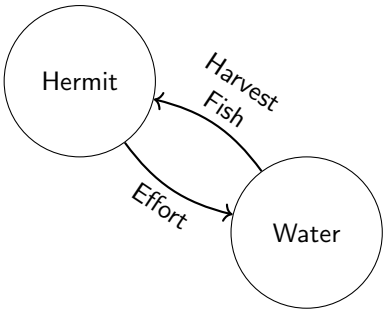
$$AP = TP/E$$

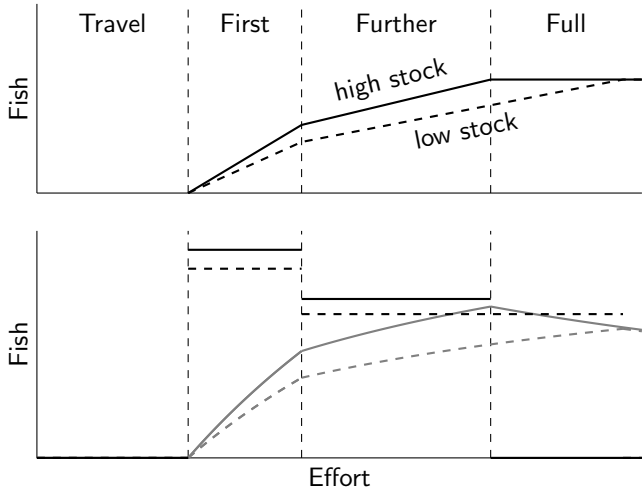
  - E can be any input, not just effort.
- Marginal Product (MP) is based on a small change in in the level of an input, around a specific level of that input.

$$MP = \Delta TP/\Delta E = \partial TP/\partial E$$

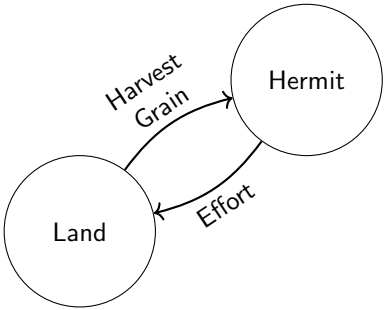
  - The marginal product measures how much extra output would be produced if a small amount of extra input is used.
  - It is the slope of the production function at the input level.
  - If production function is curved, slope of tangency line.

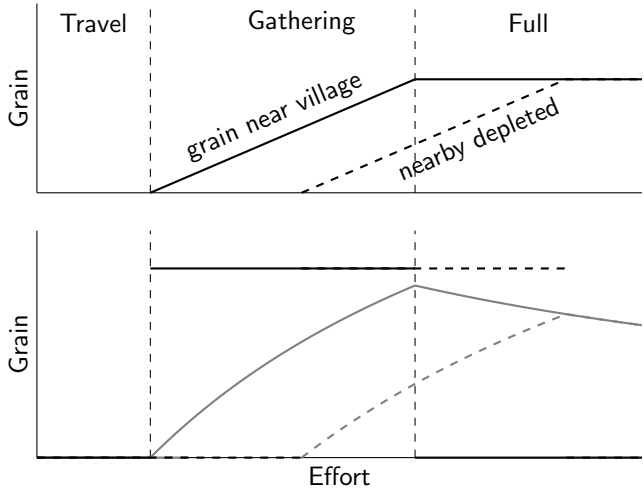
The Fish Subsystem





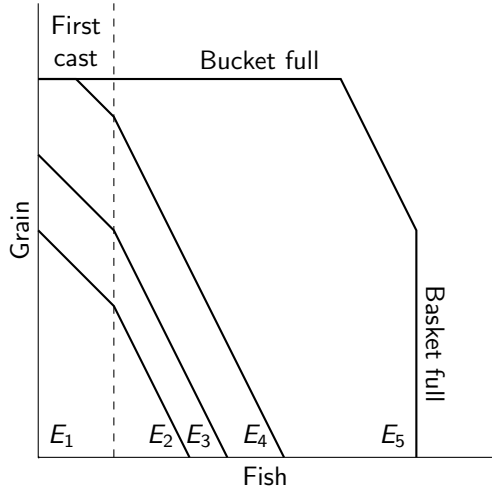
The Grain Subsystem





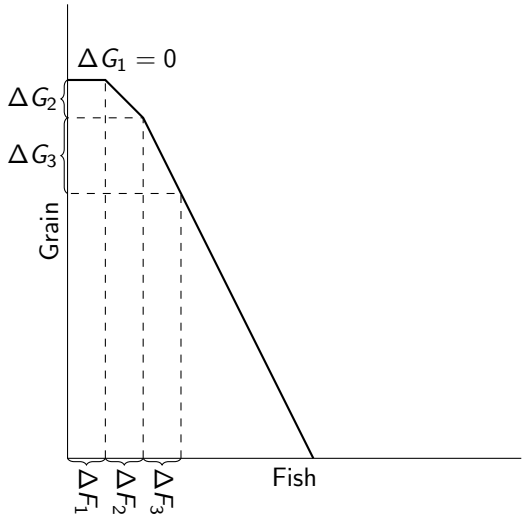
Production Possibilities

- Split total effort between fish and grain.
- $E_5 > E_4 > E_3 > E_2 > E_1$
- At E_1, no fish caught or grain gathered.
- At E_5, most effort not used, as bucket or basket full.



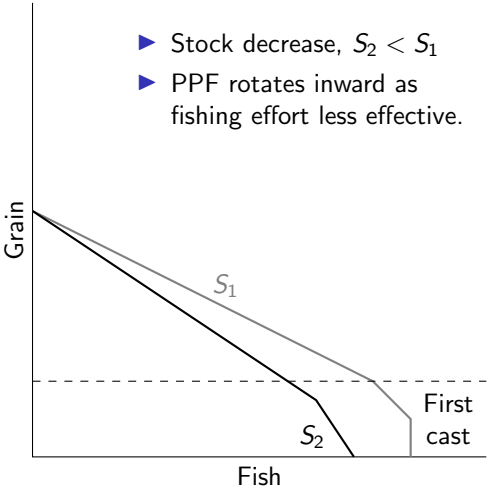
Opportunity Cost

- Increase fish by ΔF_1 for no reduction in grain.
- Increase fish by ΔF_2 for reduction in grain of ΔG_2.
- Increase fish by ΔF_3 for reduction in grain of ΔG_3.

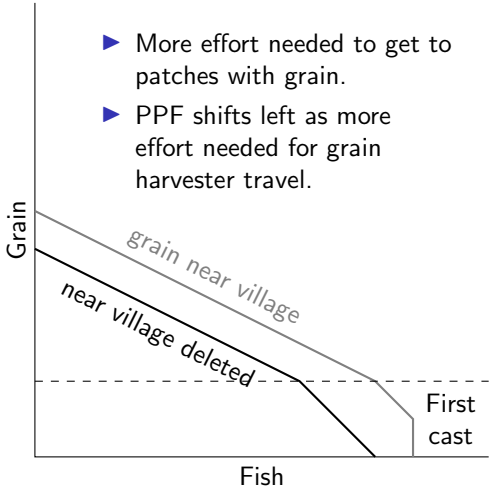


Production Possibilities

- Stock decrease, $S_2 < S_1$
- PPF rotates inward as fishing effort less effective.



- More effort needed to get to patches with grain.
- PPF shifts left as more effort needed for grain harvester travel.



Production Possibilities

- Lifespans L_1 and L_2 are attainable.
  - At **b** attain L_1.
  - At **a** attain L_2.
  - Same effort level.
- Lifespans L_3, L_4 and L_5 are not attainable.

